

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                 |
|---------------------------|---------------------------------|
| Product Description:      | <b><u>o-Toluidine</u></b>       |
| Cat No. :                 | <b>L05060</b>                   |
| Synonyms                  | 2-Aminotoluene; 2-Methylaniline |
| Index No                  | 612-091-00-X                    |
| CAS No                    | 95-53-4                         |
| Molecular Formula         | C7 H9 N                         |
| REACH registration number | -                               |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

|                                    |                    |
|------------------------------------|--------------------|
| Acute oral toxicity                | Category 3 (H301)  |
| Acute dermal toxicity              | Category 4 (H312)  |
| Acute Inhalation Toxicity - Vapors | Category 3 (H331)  |
| Skin Corrosion/Irritation          | Category 2 (H315)  |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)  |
| Germ Cell Mutagenicity             | Category 2 (H341)  |
| Carcinogenicity                    | Category 1B (H350) |

### Environmental hazards

|                          |                   |
|--------------------------|-------------------|
| Acute aquatic toxicity   | Category 1 (H400) |
| Chronic aquatic toxicity | Category 2 (H411) |

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H312 - Harmful in contact with skin  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H400 - Very toxic to aquatic life  
H411 - Toxic to aquatic life with long lasting effects  
H301 + H331 - Toxic if swallowed or if inhaled  
Combustible liquid

### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

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Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component              | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                                         |
|------------------------|---------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | 95-53-4 | EEC No. 202-429-0 | <= 100   | Acute Tox. 3 (H301)<br>Acute Tox. 4 (H312)<br>Skin Irrit. 2 (H315)<br>Eye Dam. 1 (H318)<br>Acute Tox. 3 (H331)<br>Muta. 2 (H341)<br>Carc. 1B (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 2 (H411) |

| Component              | Specific concentration limits (SCL's) | M-Factor | Component notes |
|------------------------|---------------------------------------|----------|-----------------|
| Benzenamine, 2-methyl- | -                                     | 1        | -               |

| REACH registration number | - |
|---------------------------|---|
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. Causes severe eye damage. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

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## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Combustible material. Flammable. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Protect from

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direct sunlight. Keep locked up. Keep under nitrogen. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1C

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component              | European Union                                               | The United Kingdom                                                                                                                   | France                                                                                 | Belgium                                                        | Spain                                                                                    |
|------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | TWA: 0.5 mg/m <sup>3</sup> (8h)<br>TWA: 0.1 ppm (8h)<br>Skin | STEL: 0.3 ppm 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 ppm 8 hr<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | TWA / VME: 0.1 ppm (8 heures).<br>TWA / VME: 0.5 mg/m <sup>3</sup> (8 heures).<br>Peau | TWA: 2 ppm 8 uren<br>TWA: 8.9 mg/m <sup>3</sup> 8 uren<br>Huid | TWA / VLA-ED: 0.1 ppm (8 horas)<br>TWA / VLA-ED: 0.5 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component              | Italy                                                                                                               | Germany                                                                                  | Portugal                                                           | The Netherlands                             | Finland                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | TWA: 0.5 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>TWA: 0.1 ppm 8 ore.<br>Time Weighted Average<br>Pelle | TWA: 0.1 ppm (8 Stunden). AGW -<br>TWA: 0.5 mg/m <sup>3</sup> (8 Stunden). AGW -<br>Haut | TWA: 0.1 ppm 8 horas<br>TWA: 0.5 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>TWA: 0.084 mg/m <sup>3</sup> 8 uren | TWA: 0.5 mg/m <sup>3</sup> 8 tunteina<br>TWA: 0.1 ppm 8 tunteina<br>Iho |

| Component              | Austria                                                                                                                                | Denmark                                                                                                                                 | Switzerland                                                                 | Poland                                 | Norway                                                                                                                                                                         |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | TRK-KZGW: 0.4 ppm 15 Minuten<br>TRK-KZGW: 2 mg/m <sup>3</sup> 15 Minuten<br>Haut<br>TRK-TMW: 0.1 ppm<br>TRK-TMW: 0.5 mg/m <sup>3</sup> | TWA: 0.1 ppm 8 timer<br>TWA: 0.5 mg/m <sup>3</sup> 8 timer<br>STEL: 0.2 ppm 15 minutter<br>STEL: 1 mg/m <sup>3</sup> 15 minutter<br>Hud | Haut/Peau<br>TWA: 0.1 ppm 8 Stunden<br>TWA: 0.5 mg/m <sup>3</sup> 8 Stunden | TWA: 0.5 mg/m <sup>3</sup> 8 godzinach | TWA: 0.1 ppm 8 timer<br>TWA: 0.5 mg/m <sup>3</sup> 8 timer<br>STEL: 0.3 ppm 15 minutter. value calculated<br>STEL: 0.15 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |

| Component              | Bulgaria                                                    | Croatia                                                                        | Ireland                                                                                                                      | Cyprus                                                                                | Czech Republic                                                                                                 |
|------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 0.1 ppm<br>Skin notation | kože<br>TWA-GVI: 0.1 ppm 8 satima.<br>TWA-GVI: 0.5 mg/m <sup>3</sup> 8 satima. | TWA: 0.1 ppm 8 hr.<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.3 ppm 15 min<br>STEL: 1.5 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>TWA: 0.5 mg/m <sup>3</sup><br>TWA: 0.1 ppm | TWA: 0.5 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 0.5 mg/m <sup>3</sup> |

| Component              | Estonia                                                                    | Gibraltar | Greece                                                                                  | Hungary                                                                        | Iceland                                                                                                         |
|------------------------|----------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | Nahk<br>TWA: 0.5 mg/m <sup>3</sup> 8 tundides.<br>TWA: 0.1 ppm 8 tundides. |           | skin - potential for cutaneous absorption<br>TWA: 0.1 ppm<br>TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> 8 órában. AK lehetséges borón keresztül felszívódás | TWA: 0.1 ppm 8 klukkustundum.<br>TWA: 0.5 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 4 ppm |

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|  |  |  |  |  |                               |
|--|--|--|--|--|-------------------------------|
|  |  |  |  |  | Ceiling: 18 mg/m <sup>3</sup> |
|--|--|--|--|--|-------------------------------|

| Component              | Latvia                                                                                | Lithuania                                                   | Luxembourg | Malta | Romania                                                                                |
|------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|------------|-------|----------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | skin - potential for cutaneous exposure<br>TWA: 0.5 mg/m <sup>3</sup><br>TWA: 0.1 ppm | TWA: 0.5 mg/m <sup>3</sup> IPRD<br>TWA: 0.1 ppm IPRD<br>Oda |            |       | Skin notation<br>TWA: 3 mg/m <sup>3</sup> 8 ore<br>STEL: 5 mg/m <sup>3</sup> 15 minute |

| Component              | Russia                                                                       | Slovak Republic                                                                                                           | Slovenia                                                         | Sweden                                                                           | Turkey |
|------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|
| Benzenamine, 2-methyl- | TWA: 0.5 mg/m <sup>3</sup> 0102<br>Skin notation<br>MAC: 1 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> 8<br>hodinách<br>Potential for cutaneous absorption<br>STEL: 2.5 mg/m <sup>3</sup> 15 minútach | TWA: 0.5 mg/m <sup>3</sup> 8 urah<br>TWA: 0.1 ppm 8 urah<br>Koža | TLV: 0.1 ppm 8 timmar.<br>NGV<br>TLV: 0.5 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud |        |

## Biological limit values

List source(s):

| Component              | European Union | United Kingdom | France                                                                | Spain                                                                              | Germany |
|------------------------|----------------|----------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|---------|
| Benzenamine, 2-methyl- |                |                | Methemoglobin: 1.5 %<br>of hemoglobin blood<br>during or end of shift | Methemoglobin: 1.5 %<br>Methemoglobin in total<br>hemoglobin blood end<br>of shift |         |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

## Predicted No Effect Concentration (PNEC)

See values below.

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Long sleeved clothing.                                                                                                                                                                                                                                                                                                      |
| Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination. |                                                                                                                                                                                                                                                                                                                             |
| <b>Respiratory Protection</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 Ammonia and organic ammonia derivatives filter Type K Green conforming to EN14387           |
| <b>Small scale/Laboratory use</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                      |                                          |
|------------------------------------------------|--------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                               |                                          |
| <b>Appearance</b>                              | Light yellow                         |                                          |
| <b>Odor</b>                                    | Rotten-egg like                      |                                          |
| <b>Odor Threshold</b>                          | No data available                    |                                          |
| <b>Melting Point/Range</b>                     | -23 °C / -9.4 °F                     |                                          |
| <b>Softening Point</b>                         | No data available                    |                                          |
| <b>Boiling Point/Range</b>                     | 199 - 200 °C / 390.2 - 392 °F        |                                          |
| <b>Flammability (liquid)</b>                   | Combustible liquid                   | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                       | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.5<br><b>Upper</b> 7.5 |                                          |
| <b>Flash Point</b>                             | 85 °C / 185 °F                       | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 480 °C / 896 °F                      |                                          |
| <b>Decomposition Temperature</b>               | > 200°C                              |                                          |
| <b>pH</b>                                      | 7.5                                  | 8 g/l aq. sol                            |
| <b>Viscosity</b>                               | No data available                    |                                          |
| <b>Water Solubility</b>                        | 1.5 G/100ML (25°C)                   |                                          |
| <b>Solubility in other solvents</b>            | No information available             |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                      |                                          |
| <b>Component</b>                               | <b>log Pow</b>                       |                                          |
| Benzenamine, 2-methyl-                         | 1.4                                  |                                          |
| <b>Vapor Pressure</b>                          | 0.3 mbar @ 20 °C                     |                                          |
| <b>Density / Specific Gravity</b>              | 1.000                                |                                          |
| <b>Bulk Density</b>                            | Not applicable                       | Liquid                                   |
| <b>Vapor Density</b>                           | 3.7 (Air = 1.0)                      | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)              |                                          |

### 9.2. Other information

|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Molecular Formula</b>    | C7 H9 N                                |
| <b>Molecular Weight</b>     | 107.15                                 |
| <b>Explosive Properties</b> | explosive air/vapour mixtures possible |

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## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions. Light sensitive. Air sensitive.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization Hazardous Reactions

No information available.  
None under normal processing.

### 10.4. Conditions to avoid

Exposure to air. Exposure to light. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 4 |
| Inhalation | Category 3 |

| Component              | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation            |
|------------------------|--------------------------|------------------------------|----------------------------|
| Benzenamine, 2-methyl- | LD50 = 670 mg/kg ( Rat ) | LD50 = 3235 mg/kg ( Rabbit ) | LC50 = 862 ppm ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component              | EU           | UK | Germany | IARC    |
|------------------------|--------------|----|---------|---------|
| Benzenamine, 2-methyl- | Carc Cat. 1B |    | Cat. 1  | Group 1 |

(g) reproductive toxicity; No data available



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(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

| Component              | Freshwater Fish | Water Flea | Freshwater Algae                                                                                          |
|------------------------|-----------------|------------|-----------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- |                 |            | EC50: = 3.7 mg/L, 96h static (Desmodesmus subspicatus)<br>EC50: = 3.9 mg/L, 72h (Desmodesmus subspicatus) |

| Component              | Microtox                | M-Factor |
|------------------------|-------------------------|----------|
| Benzenamine, 2-methyl- | EC50 = 13.2 mg/L 30 min | 1        |

### 12.2. Persistence and degradability

Persistence Persistence is unlikely.  
Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

| Component              | log Pow | Bioconcentration factor (BCF) |
|------------------------|---------|-------------------------------|
| Benzenamine, 2-methyl- | 1.4     | No data available             |

12.4. Mobility in soil The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

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**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                    |
|-----------------------------------------|--------------------|
| <u>14.1. UN number</u>                  | UN1708             |
| <u>14.2. UN proper shipping name</u>    | TOLUIDINES, LIQUID |
| <u>14.3. Transport hazard class(es)</u> | 6.1                |
| <u>14.4. Packing group</u>              | II                 |

### ADR

|                                         |                    |
|-----------------------------------------|--------------------|
| <u>14.1. UN number</u>                  | UN1708             |
| <u>14.2. UN proper shipping name</u>    | TOLUIDINES, LIQUID |
| <u>14.3. Transport hazard class(es)</u> | 6.1                |
| <u>14.4. Packing group</u>              | II                 |

### IATA

|                                         |                    |
|-----------------------------------------|--------------------|
| <u>14.1. UN number</u>                  | UN1708             |
| <u>14.2. UN proper shipping name</u>    | TOLUIDINES, LIQUID |
| <u>14.3. Transport hazard class(es)</u> | 6.1                |
| <u>14.4. Packing group</u>              | II                 |

14.5. Environmental hazards

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

14.6. Special precautions for user

No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

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## International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component              | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Benzenamine, 2-methyl- | 95-53-4 | 202-429-0 | -      | -   | X     | X    | KE-23446 | X    | X    |

| Component              | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Benzenamine, 2-methyl- | 95-53-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component              | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                  | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------|---------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | 95-53-4 | -                                                                   | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 43.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - Carcinogenic (Article 57a)                                                      |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component              | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | 95-53-4 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

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UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** See table for values

| Component              | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                            |
|------------------------|---------------------------------------|------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | WGK3                                  | Krebserzeugende Stoffe - Class II : 0.5 mg/m <sup>3</sup><br>(Massenkonzentration) |

| Component              | France - INRS (Tables of occupational diseases)                        |
|------------------------|------------------------------------------------------------------------|
| Benzenamine, 2-methyl- | Tableaux des maladies professionnelles (TMP) - RG 15,RG 15bis,RG 15ter |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed  
H312 - Harmful in contact with skin  
H331 - Toxic if inhaled  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H400 - Very toxic to aquatic life  
H411 - Toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical incident response training.

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|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Revision Date    | 10-Feb-2024                                        |
| Revision Summary | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**